

Sunil Maharana

AI/ML ENGINEER | GEN AI | AI ARCHITECT | LLMOps

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Professional Summary

AI Engineer and former Strategic Procurement Manager with 8 years' experience—5 years in supply chain/manufacturing and 3 years hands-on in GenAI, LLMs, and MLOps. Specialized in designing, deploying, and monitoring robust AI/GenAI solutions using AWS, Langchain, and modern agentic workflows. Adept at the full AI lifecycle: data versioning, modular pipeline design, CI/CD, model registry, monitoring, and retraining. Proven ability to deliver impactful AI applications and optimize business processes in real-world industrial environments.

Core Skills

GenAI & LLMs:: Langchain, LlamaIndex, OpenAI GPT-3.5/4 (API), Claude Haiku (Bedrock API), Hugging Face Transformers, RAG, Agentic AI, Prompt Engineering

MLOps & AWS:: SageMaker, Lambda, ECS, ECR, S3, IAM, VPC, CloudWatch, Prometheus, Docker, MLflow, GitHub Actions, Step Functions, Bedrock

Classic ML:: Python, Scikit-learn, XGBoost, BERT, EfficientNet, DVC, Feature Store

DevOps:: CI/CD, Modular Pipelines, FastAPI, Streamlit, Grafana, Model Registry

Domain:: Supply Chain Optimization, Procurement Analytics, Manufacturing AI

Soft Skills:: Problem-solving, Communication, Stakeholder Management, Critical Thinking

Work Experience

Sugna Metals Ltd., Hyderabad **Nov 2017 – Present** *AI Engineer (Supply Chain)*

- Led AI-driven solutions for steel quality, inventory, production delays, and quality escalations; integrated data sources and ensured secure, scalable deployments using AWS and DevOps.
- Orchestrated full lifecycle management of AI projects, including model versioning, CI/CD with GitHub Actions, and automated retraining with SageMaker and Lambda.
- Collaborated with cross-functional teams to deliver business impact, achieving cost reductions and process improvements across a 600 Cr portfolio.

Projects

RAG-Based Industrial Query Bot

Grafana

Langchain, LlamaIndex, OpenAI API, Pinecone, FastAPI, Docker, AWS ECS, Prometheus,

Architected a RAG-based QA system for internal SOPs, enabling plant engineers to retrieve actionable instructions via natural language queries. Deployed backend on AWS ECS with FastAPI, token-based access, and monitoring. Reduced information retrieval time by 80% in demo environments.

Agentic Procurement Negotiator OpenAI GPT-4 API, LangGraph, Streamlit, AWS Lambda, Qdrant, Google Sheets API Built an agentic AI app simulating multi-step procurement negotiations using LLMs and real-time data. Modular agents for offer analysis, risk, and contract drafting; deployed Streamlit frontend and Lambda backend. Achieved 30% faster negotiation cycles in demo scenarios.

GenAI-Powered Vendor Performance Analyzer Hugging Face Transformers, Streamlit, Plotly, AWS S3, MLflow, Docker Automated extraction, summarization, and visualization of vendor performance from unstructured reports using transformer models via API. Integrated Streamlit dashboard and MLflow tracking; reduced manual analysis time by 60%.

Autonomous SOP Compliance Checker OpenAI GPT-3.5 API, Langchain, FastAPI, MongoDB, Prometheus, Docker AI system parsing logs and checking compliance against SOPs, deployed as FastAPI microservice on AWS ECS. Automated compliance audit, reducing manual effort by 70%.

Production Delay Reasoning Agent BERT, Langchain, Grafana, AWS Lambda, Docker LLM-powered chatbot classifying and explaining production delays from operator logs. Integrated with Grafana for realtime visualization; improved root cause identification by 50%.

GenAI-Driven Inventory Insights Bot Llama 2 API, Langchain, Streamlit, AWS S3, Pinecone, Docker Conversational agent for inventory optimization, integrating ERP data and Pinecone vector search; reduced review time by 40%.

Resume Parser AI Project spaCy, scikit-learn, OpenAI API, FastAPI, AWS Lambda, S3, MongoDB Developed an NLP-driven resume parser to extract and structure candidate information from unstructured resumes. Used spaCy for NER, OpenAI API for skill extraction, and scikit-learn for clustering. Deployed as a FastAPI microservice on AWS Lambda, storing results in MongoDB and S3.

GenAI Tabular & Time Series Data Analysis Agent

Langchain, OpenAI API, sktime, tsfresh, Pandas, Streamlit, AWS S3,

FastAPI

Built an LLM-powered agent that analyzes, summarizes, and generates insights from tabular and time series datasets. Used `tsfresh` and `sktime` for feature extraction, Langchain for NLQ, and deployed as a Streamlit app and FastAPI microservice.

Foundational ML Projects (2020–2022)

Fraud Detection with XGBoost

Built and tuned pipeline achieving 98% precision.

Mini Food Classifier

Fine-tuned EfficientNetB2, deployed interactive demo on Hugging Face Spaces.

MLOps & Deployment Experience

- Data Versioning: DVC with S3 for datasets and model artifacts.
- Experiment Tracking: MLflow and SageMaker Experiments for hyperparameter and model tracking.
- Model Registry & Deployment: SageMaker Model Registry, Lambda triggers, ECR for containerized models, serverless endpoints for cost efficiency.
- Monitoring & Retraining: Model Monitor, Prometheus, CloudWatch, event-driven retraining using Lambda and S3 triggers.
- CI/CD: GitHub Actions for automated build, test, and deployment pipelines; integrated with AWS CLI/SDK.
- Security & Networking: IAM roles, VPC, security groups, private subnets for secure and scalable deployments.

Education

Osmania University, Hyderabad

2018 – 2022

B.Tech in Computer Science

Certifications

Machine Learning with Python (IBM, 2025)

Data Science Methodologies (IBM, 2025)

Python for Data Science (IBM, 2025)